EXPERIMENT NO. – 10

Code:

```
import numpy as np
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# Input Data (Days)
X = np.array([[1], [2], [3], [4], [5]], dtype=float)

# Output Data (Sales)
y = np.array([[100], [120], [130], [150], [170]], dtype=float)

# Normalize data
X = X / 5
y = y / 200

# Create Deep Learning Model
model = Sequential()

# Hidden Layer
model.add(Dense(8, input_dim=1, activation='relu'))

# Output Layer
model.add(Dense(1, activation='linear'))

# Compile Model
model.compile(optimizer='adam',
        loss='mean_squared_error')

# Train Model using Backpropagation
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```
model.fit(X, y, epochs=500, verbose=0)


# Predict Future Sales
prediction = model.predict(np.array([[6]]) / 5)


# Display Output
print("Predicted Sales for Day 6:")
print(prediction * 200)
```

Output: